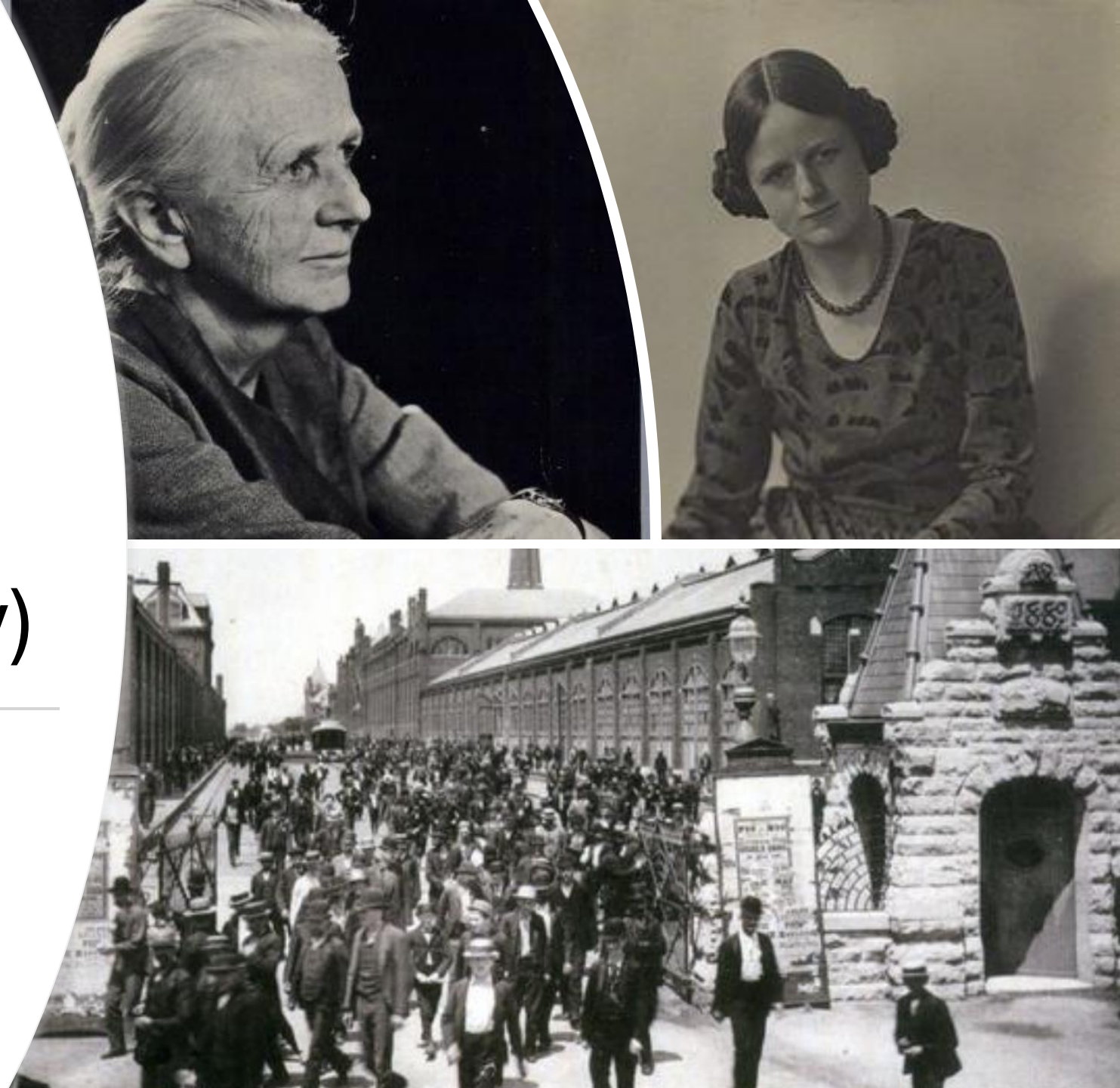


Econ 139: Labor Economics

Monopsony (Theory)

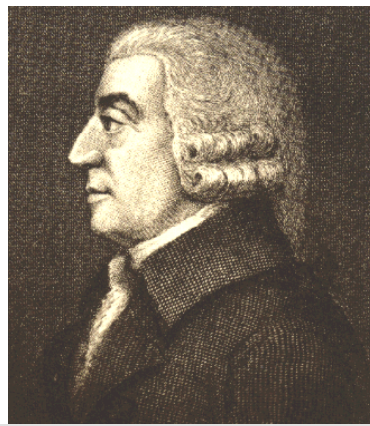
David Arnold



Adam Smith

- Adam Smith is widely known as the “father of economics”, popularizing a concept known as the “invisible hand” which leads perfectly competitive markets to welfare-maximizing outcomes
- But what is less discussed is that he also mentioned some settings in which he thought the conditions for perfect competition were not met
- *What are the common wages of labor, depends everywhere upon the contract usually made between [employers and employees], whose interests are by no means the same... The masters, being fewer in number, can combine much more easily ... Masters are always and everywhere in a sort of tacit, but constant and uniform, combination, not to raise the wages of labor above their actual rate.*

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Joan Robinson



- This aspect of labor markets was more-or-less ignored until Joan Robinson's book *The Economics of Imperfect Competition*
- This was a formal analysis of **monopsony power** which is the equivalent to market power in product markets, but on input markets instead
 - Labor was a key input market discussed
- But Robinson herself in 1969 said, "All this had no effect. Perfect competition, supply and demand...and marginal products still reign supreme in orthodox teaching"

The 20th Century

- Throughout much of the 20th century, perfect competition was the dominant paradigm through which economists analyzed labor markets
- Some associate monopsony power with literally a single employer, such as a company town, which doesn't match reality
 - But monopsony power can exist even if there are many firms in the market
 - Do you want to work for every single firm? Do you think people can immediately find a job just as good as their current job?

So What is Monopsony Power?

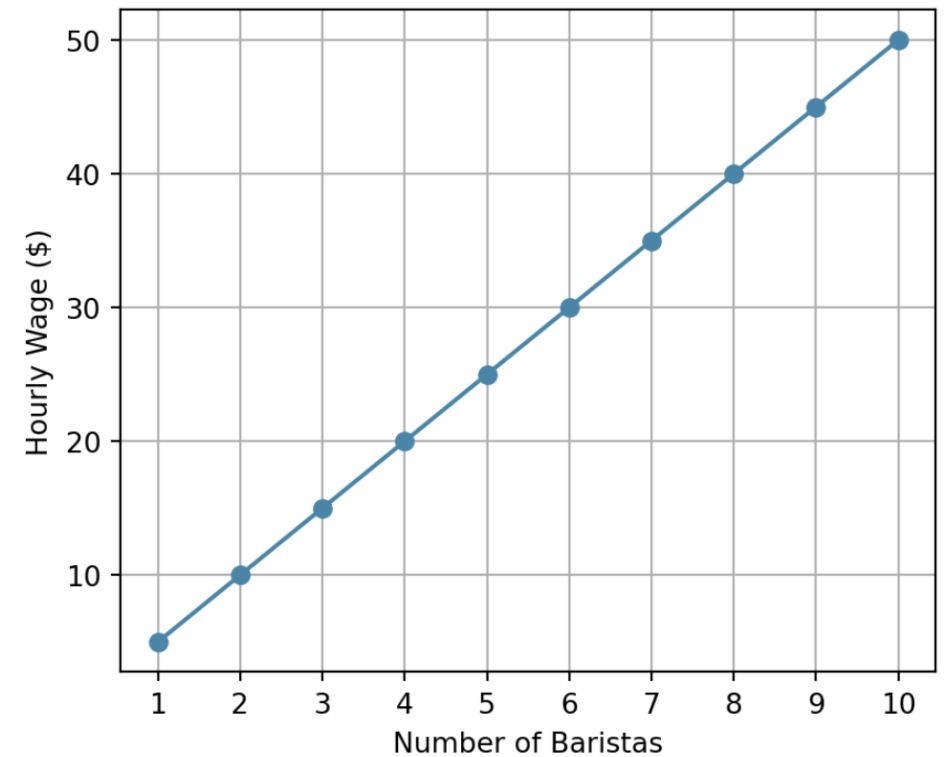
- For the perfectly competitive model, we assumed there was a market wage and if you pay even a cent beneath that wage you lose all your workers
- In this section we will assume that firms face a labor supply curve which we will denote $L(w)$
 - We will assume this is upward sloping (i.e. if I want to hire more workers, I need to pay a higher wage. If I lower my wage, I lose some of my workers but not all of them).
- This seems like a pretty minor assumption, but if you believe a firm faces an upward sloping labor supply curve, then you believe in monopsony power

So What is Monopsony Power?

- It's important to pause how little we've changed from the perfectly competitive model
 - We have made no stance on how many employers there are in the market (i.e. we are not assuming a company town)
- All we've said is that we assume if a firm wants to hire more workers, they must pay a higher wage, and that if they lower their wage, they won't lose all their workers
- If anything, a priori, this seems much more realistic than the perfectly competitive implications (i.e. I pay a cent below market wage I attract no workers).

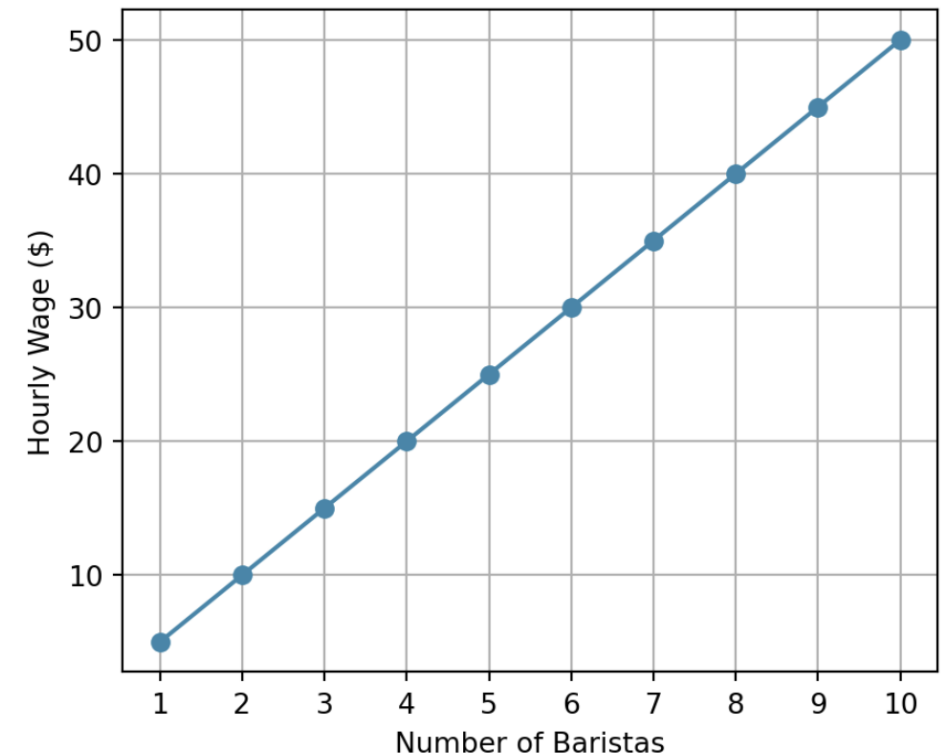
An Upward Sloping Labor Supply Curve

- Let's go back to our original example
- We are a café owner, trying to hire baristas
- Now, we assume the café faces an upward sloping labor supply curve seen here



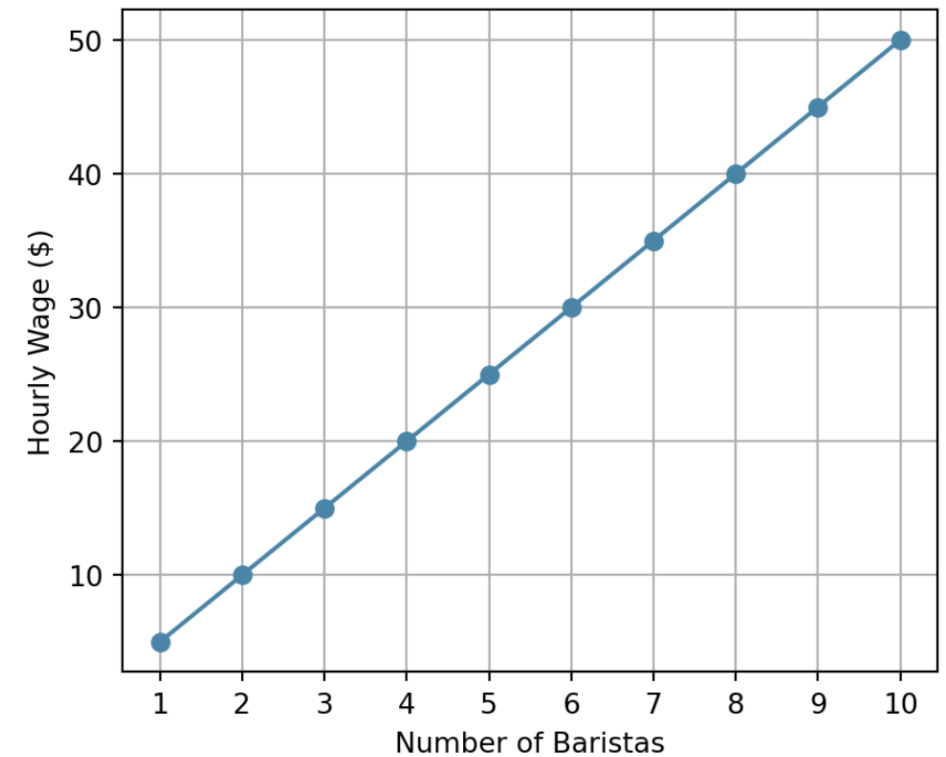
An Upward Sloping Labor Supply Curve

- How many workers should the café hire?
- To hire the first worker, needs to pay \$5
- Assume worker makes 13 lattes per hour, which sell for \$5, implying $MRPL = \$65$



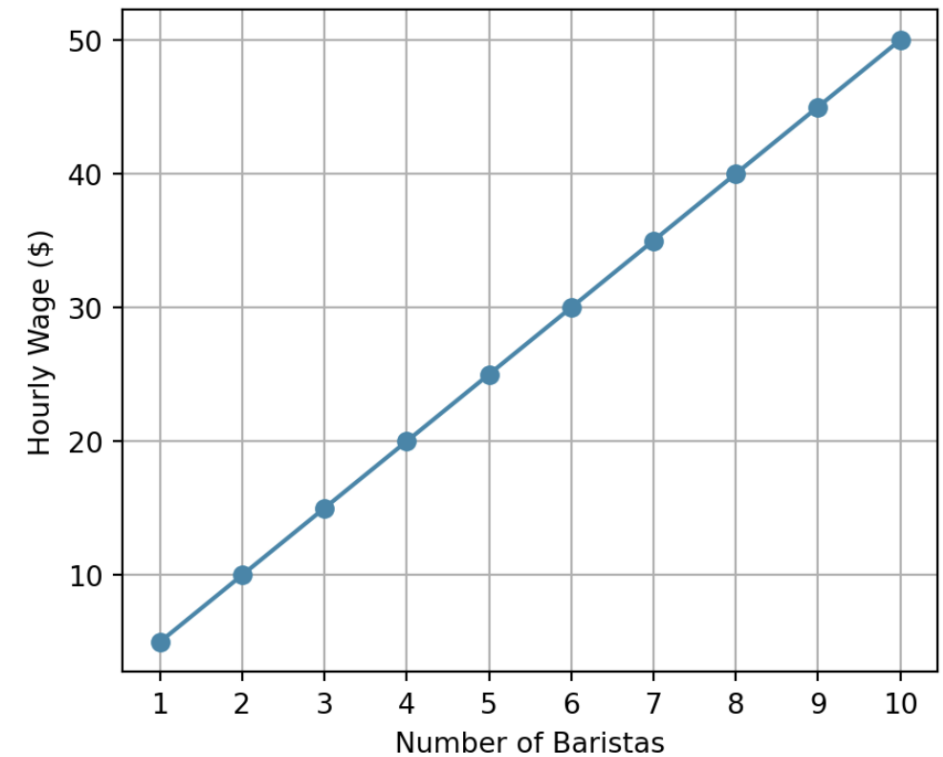
An Upward Sloping Labor Supply Curve

- Should the firm hire a 2nd worker?
- Assume worker makes 12 additional lattes per hour, which sell for \$5, implying $MRPL = \$60$
- What do we compare \$60 to?



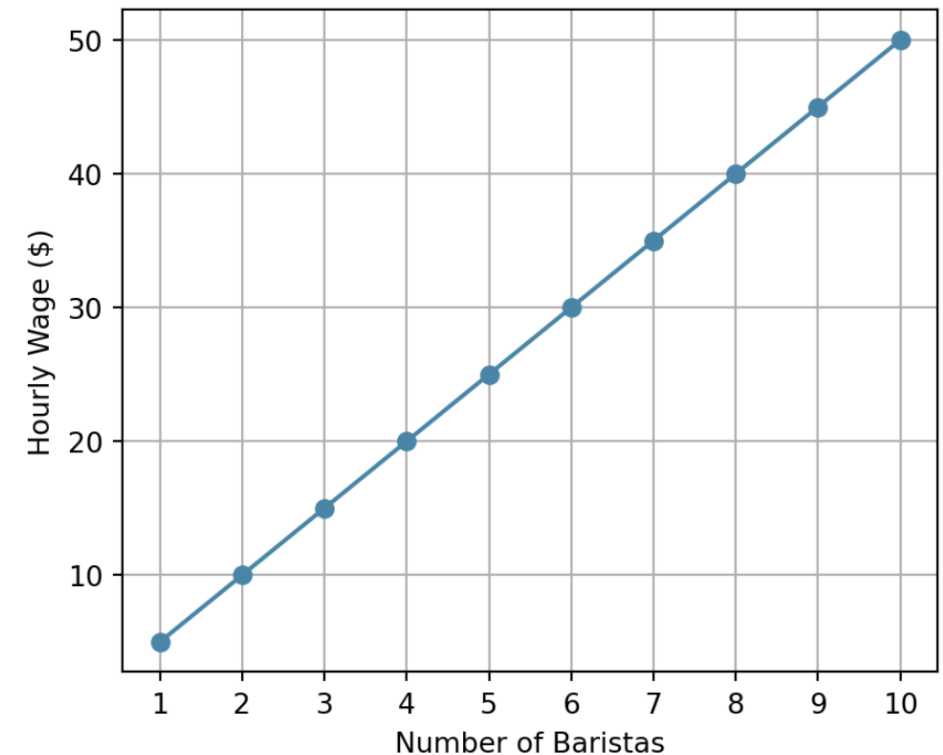
Marginal Factor Cost

- What is the cost of hiring a 2nd worker?
- \$10 seems like an obvious answer, but that's not quite right
- To hire 2 workers, you need to pay \$10 to **both** workers, so your total wage bill increases from 5 to 20



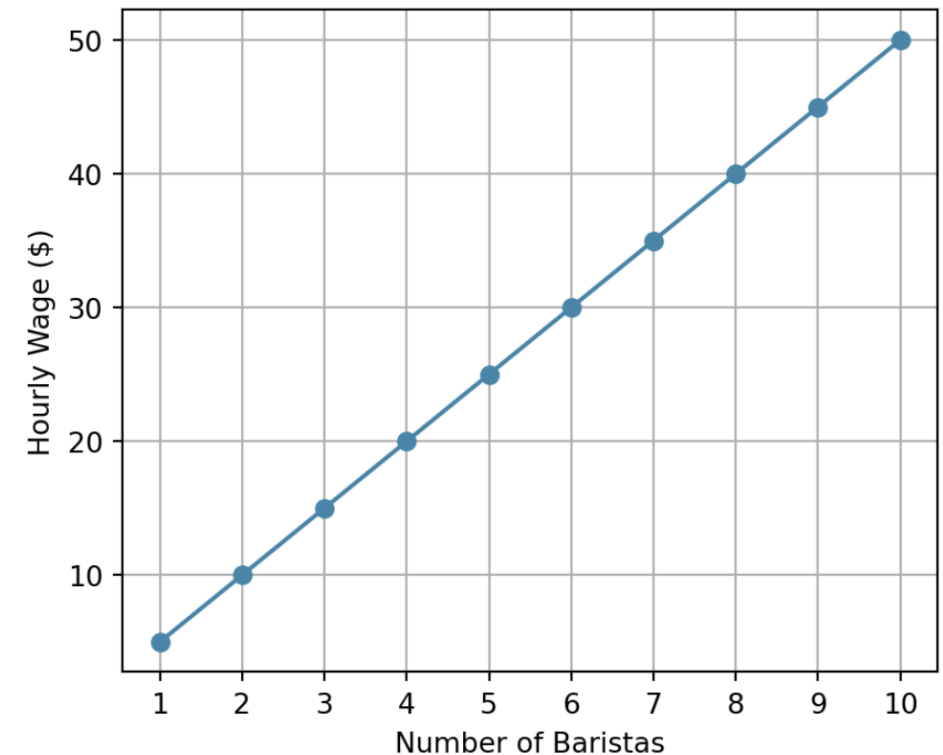
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Inframarginal Cost

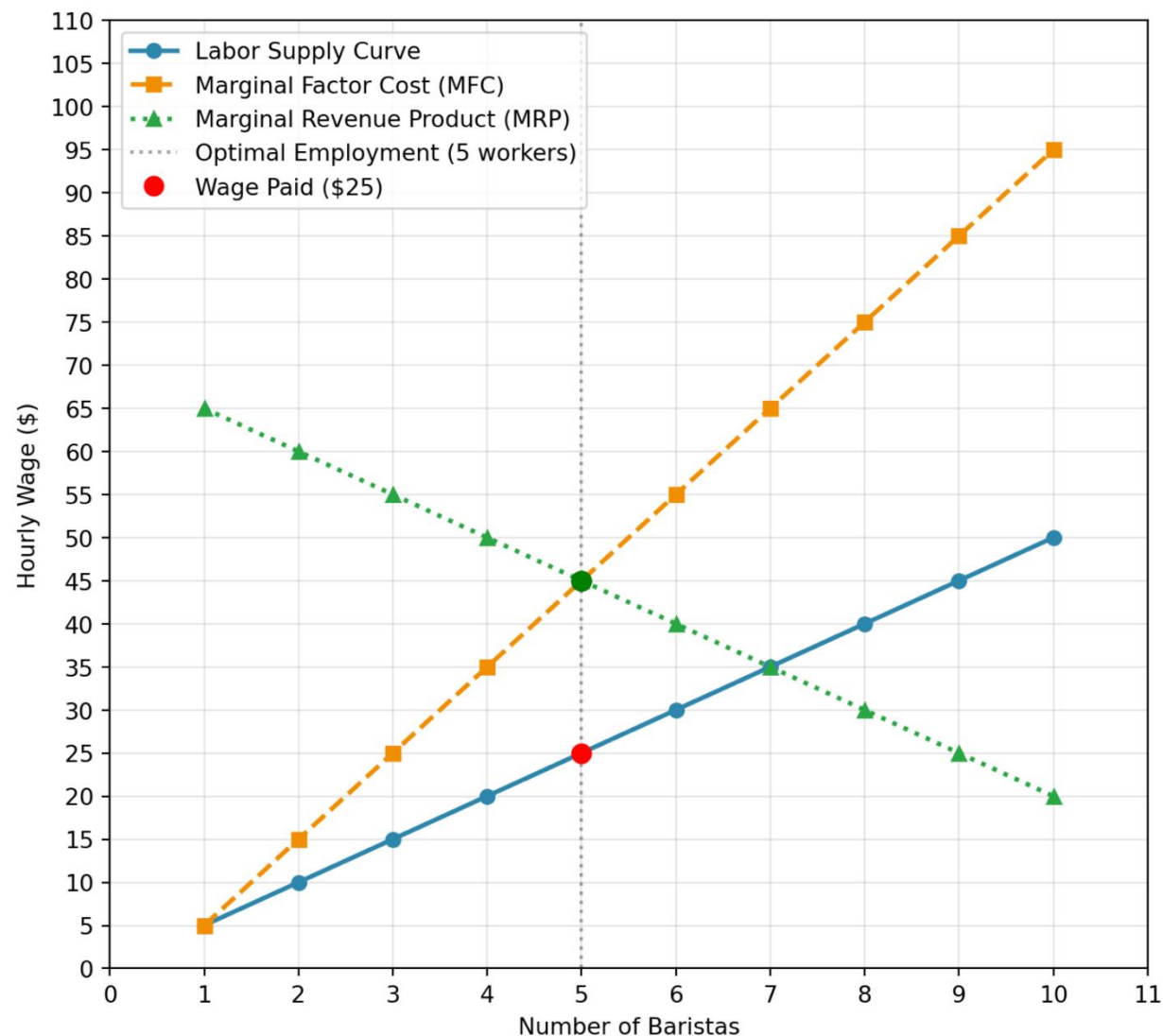
- When you hire a second worker, you face an **inframarginal cost**
- This is the cost associated with the increase in the wage for the worker you were already attracting at a wage of \$5
- In the perfectly competitive model, the marginal factor cost (the cost of an additional worker) is equal to the wage because we assumed there is a market wage and you could hire as many workers as you like at that wage

Inframarginal Cost

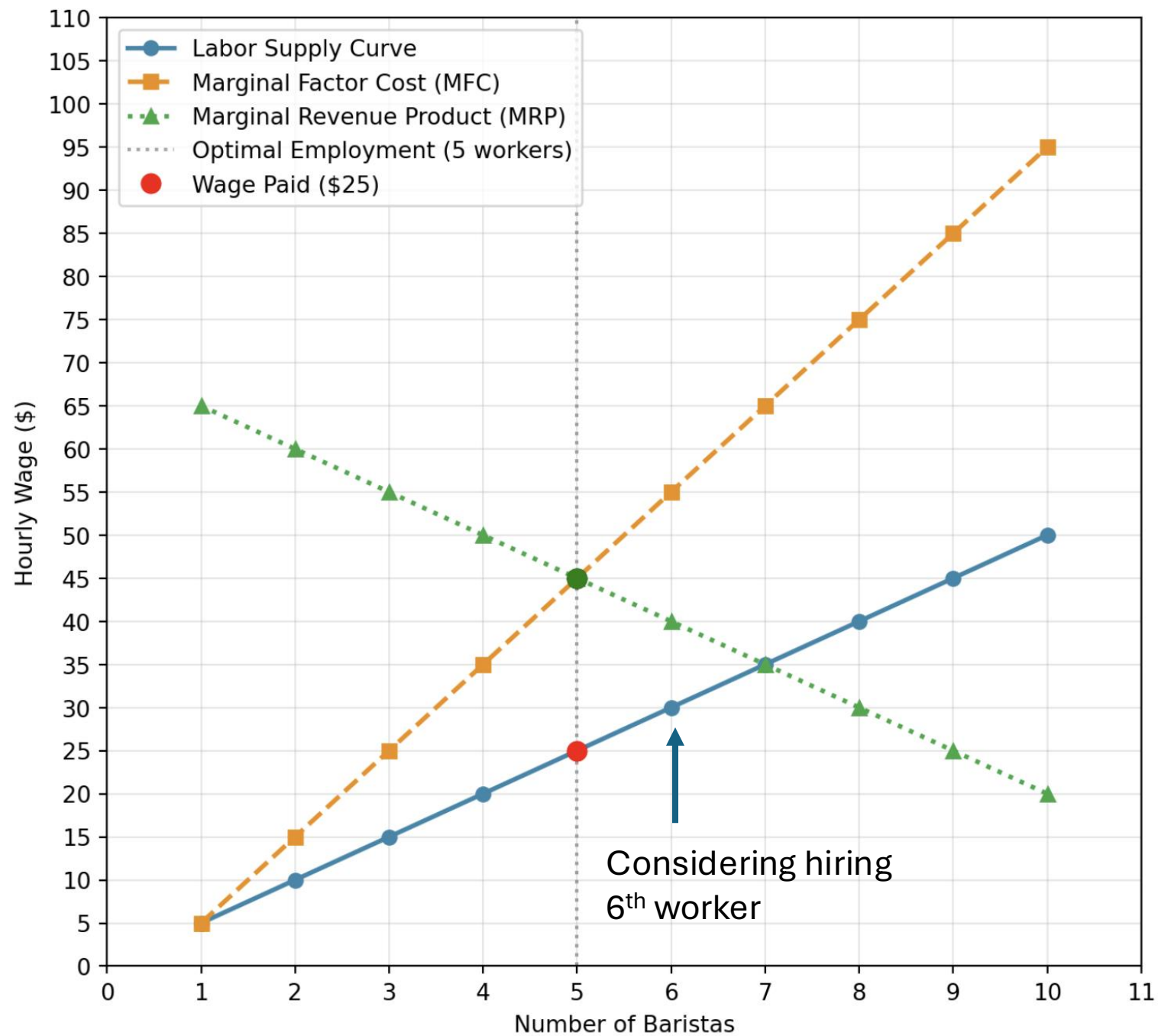
- Whether you should hire an additional worker depends on whether the **marginal factor cost** is smaller than the **marginal revenue product of labor**
- We can compute the marginal factor cost and add it to our figure, along with the marginal revenue product of labor

Wage and Employment in Monopsony

- Marginal revenue product of labor equals marginal factor cost at 5 workers
- So firm will hire 5 workers and pay a wage of \$25

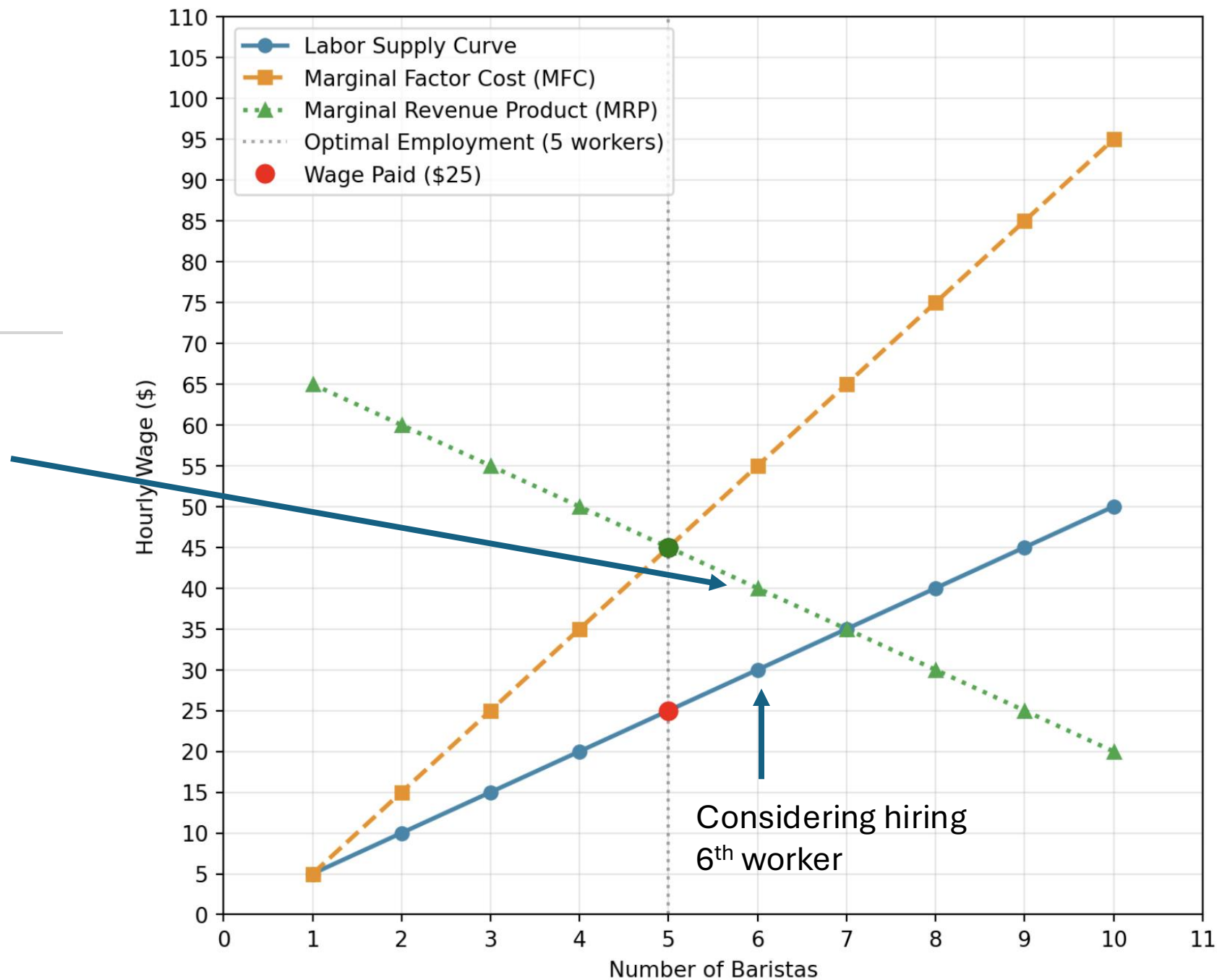


Inefficiency



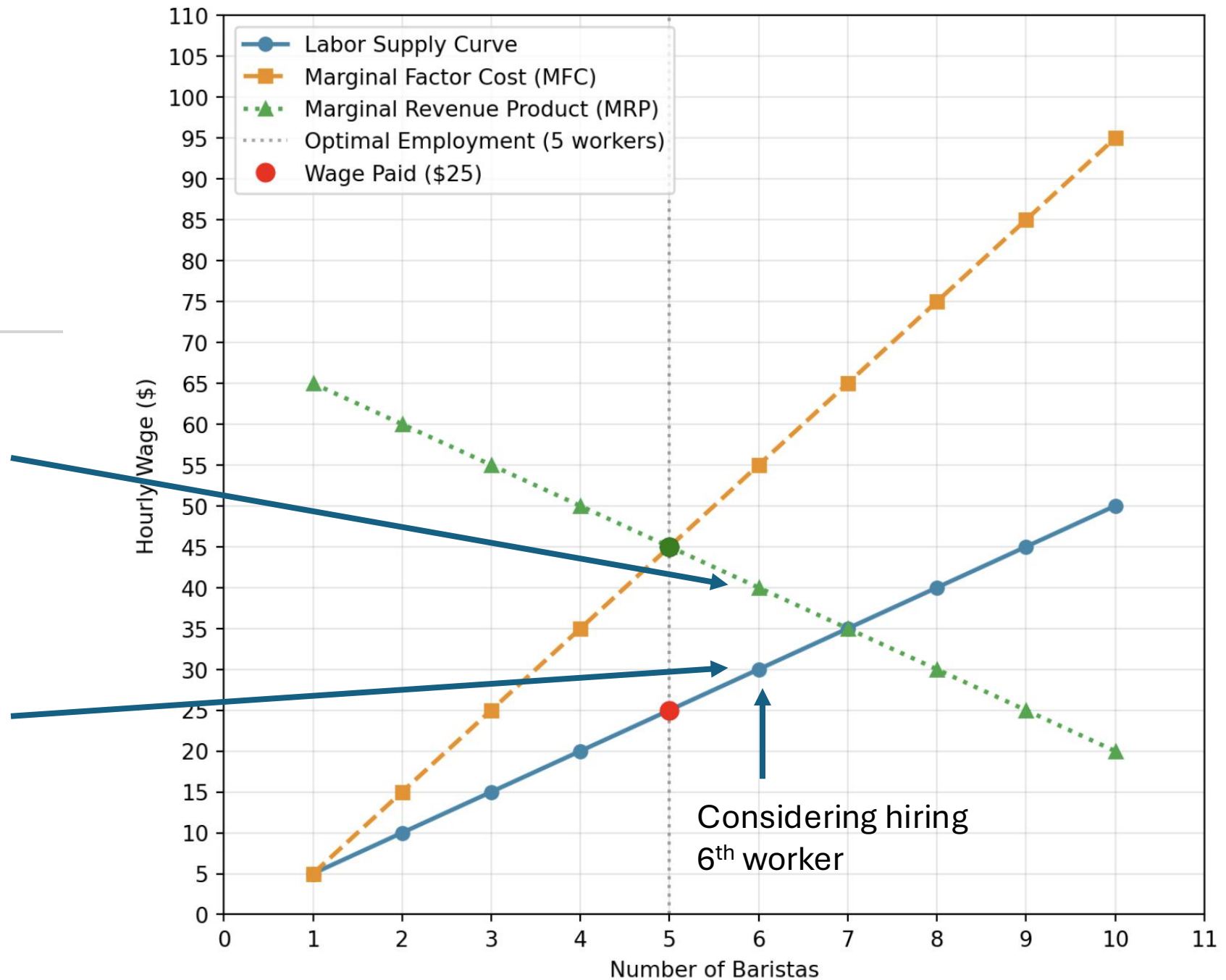
Inefficiency

- Firm makes \$40 from 6th worker
- Worker would work for \$30



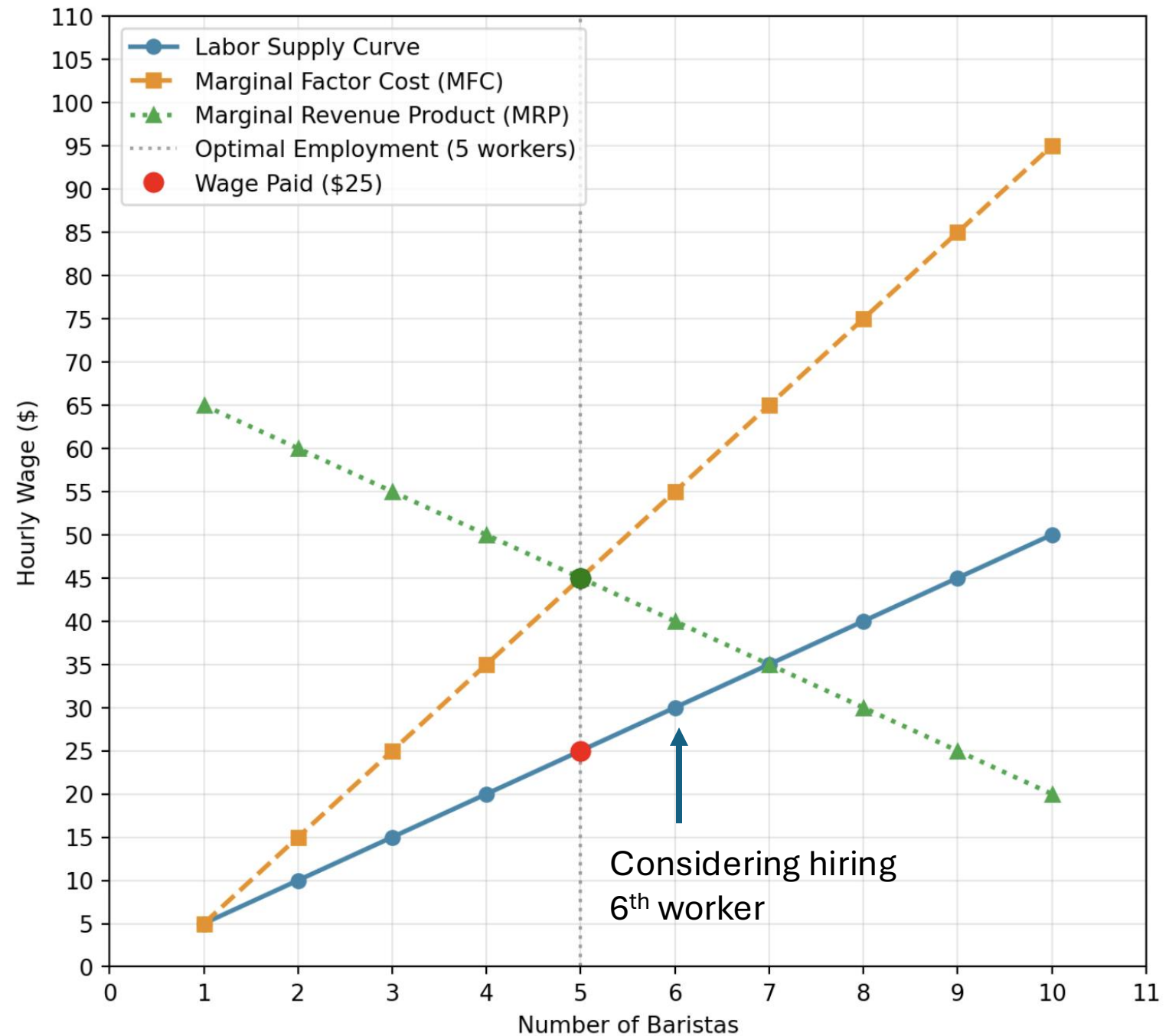
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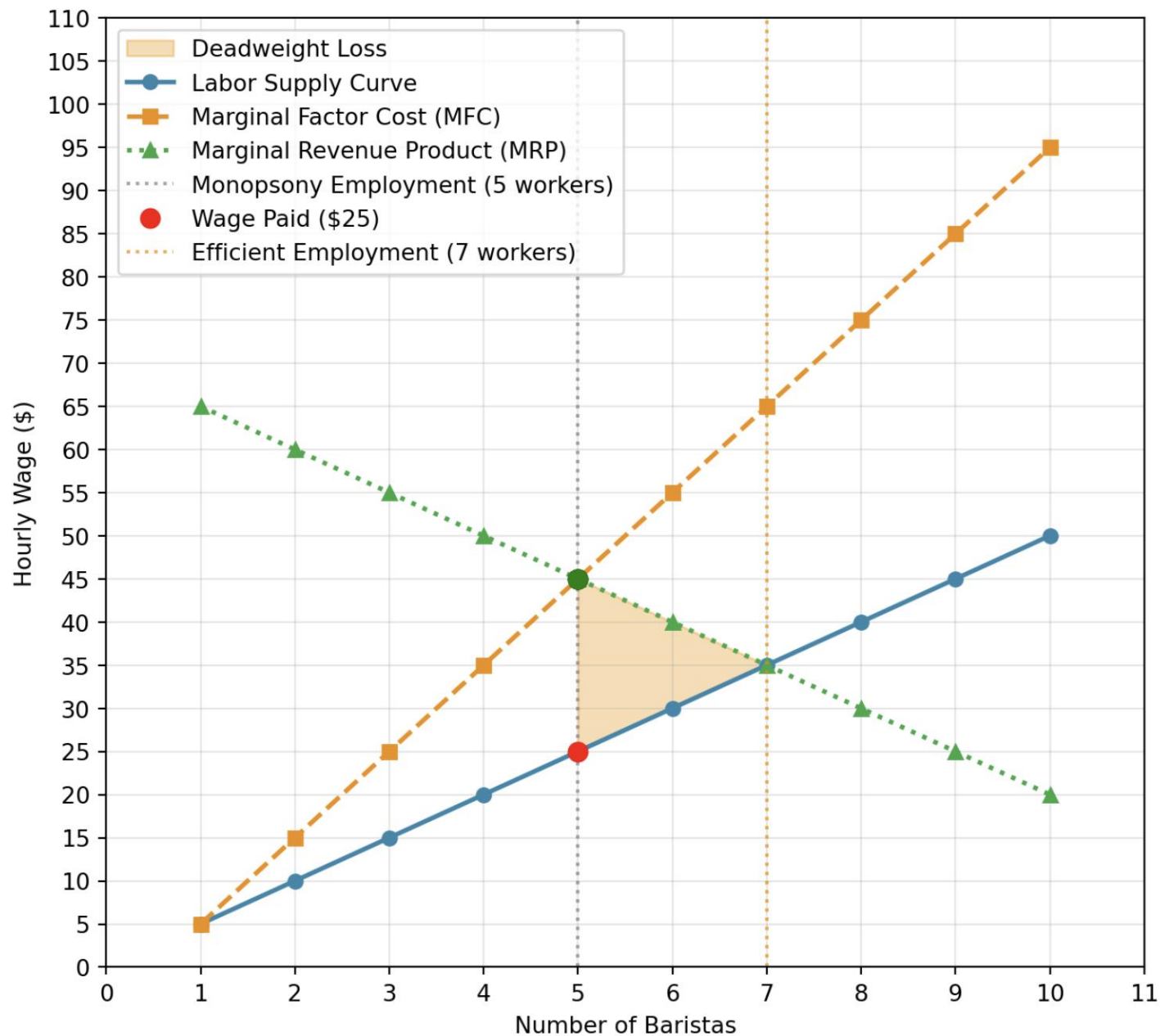
Inefficiency

- Trade doesn't occur due to inframarginal costs

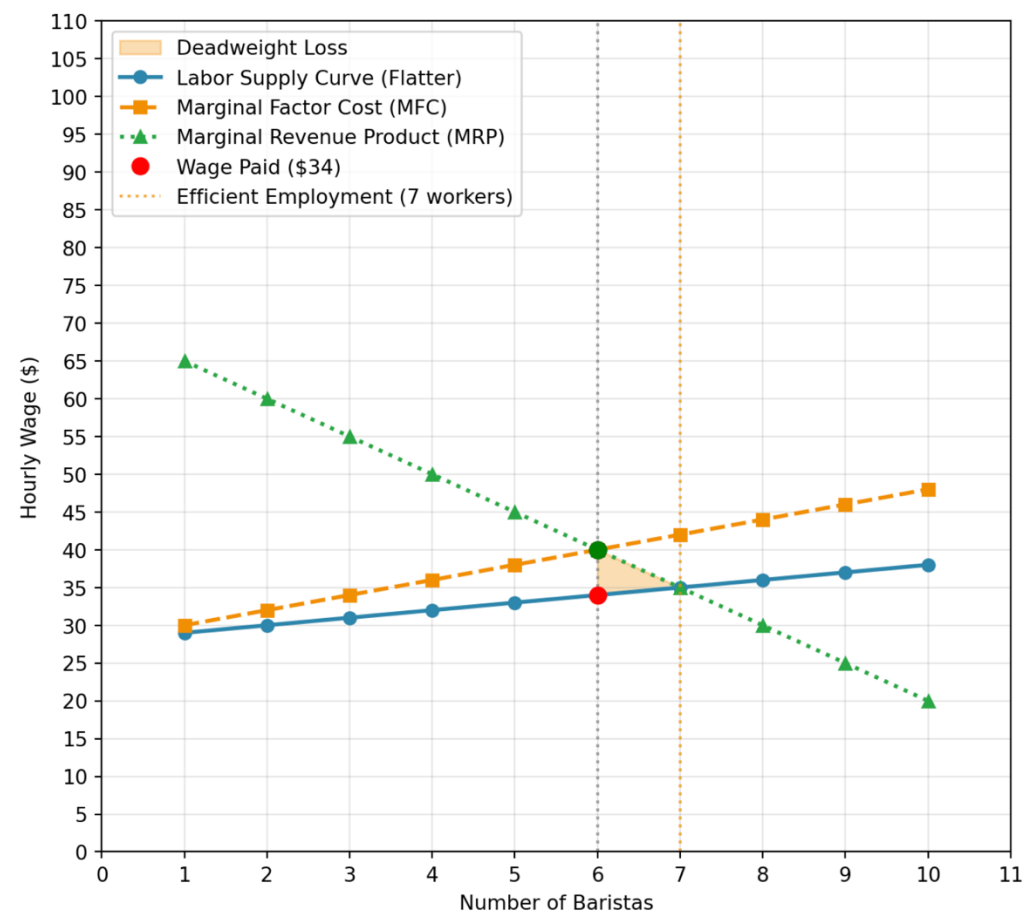
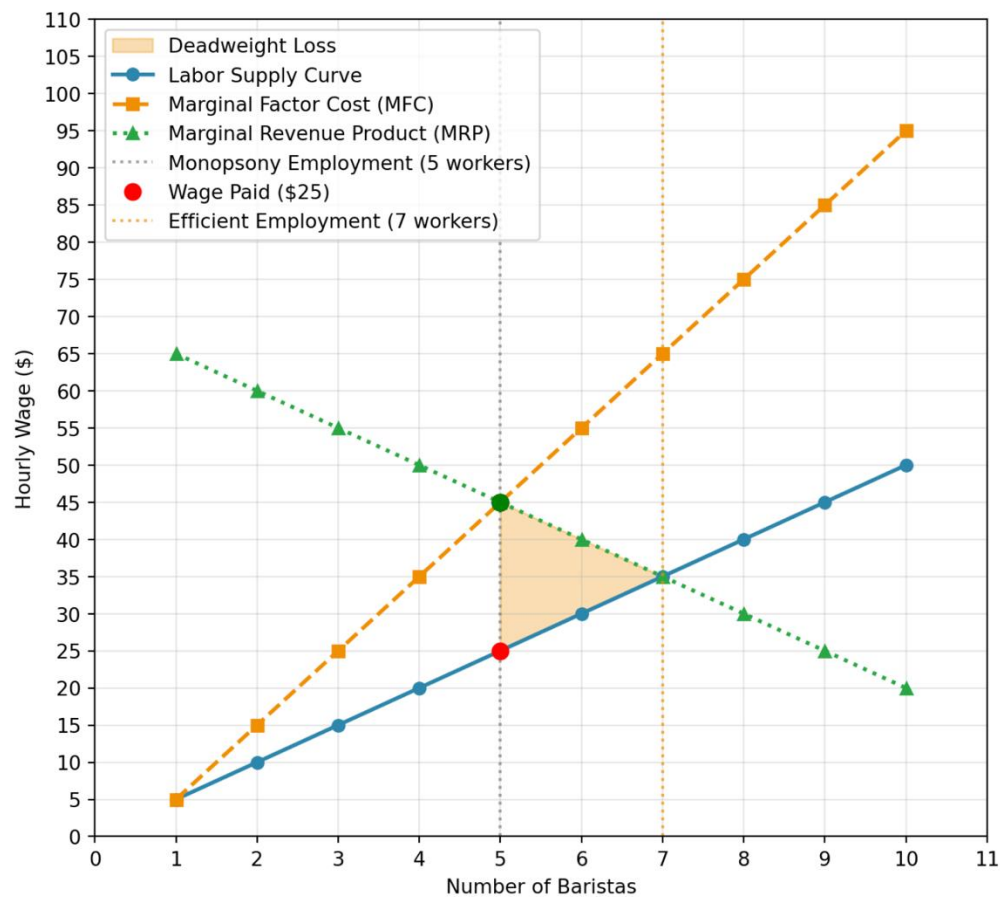


Inefficiency

- Trade doesn't occur due to inframarginal costs
- If you could hire “fraction of worker” every worker between 5 and 7 creates deadweight loss



Deadweight Loss Depends on Labor Supply Curve



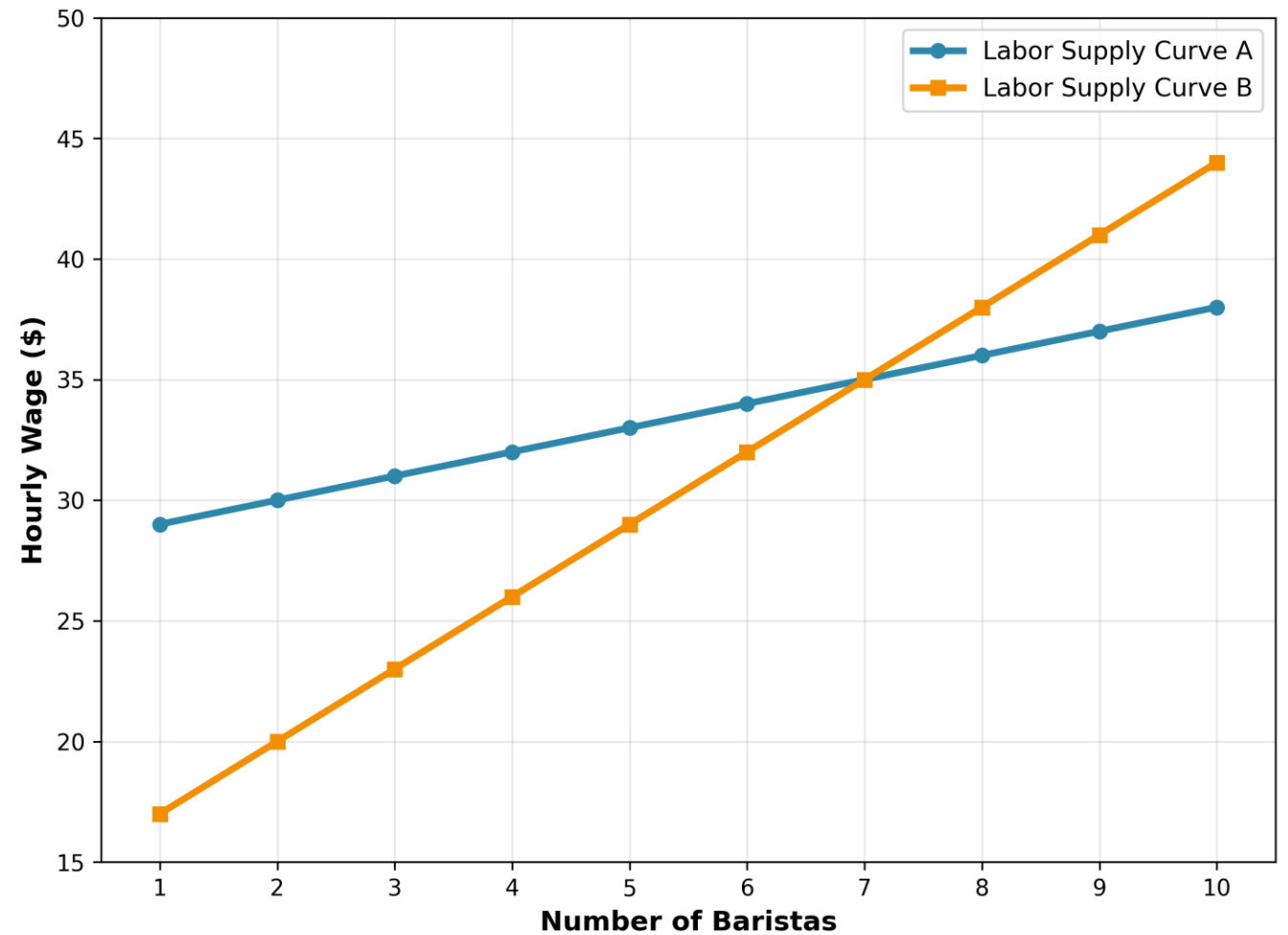
The Elasticity of Labor Supply

- The key parameter that determines the deadweight loss is the elasticity of labor supply at the firm level

$$\eta = \frac{\Delta L / L}{\Delta w / w} = \frac{\% \Delta L}{\% \Delta w}$$

- **Perfect Competition:** η is equal to infinity (i.e. I lower my wage even a miniscule amount, I lose all my workers)
- **Monopsony power:** η is finite. The smaller the value, the more scope for monopsony power. The more inelastic, the higher the scope for monopsony power

Which Labor
Supply Curve is
More Elastic



Measuring Monopsony Power

- Large literature has estimate elasticity of labor supply at the firm level to measure monopsony power

$$\eta = \frac{\Delta L / L}{\Delta w / w} = \frac{\% \Delta L}{\% \Delta w}$$

- We will discuss these in the next portion of the class, and why it is so difficult
- Another option is to try and measure the markdown

$$\text{Wage Markdown} = \frac{MRPL - W}{W}$$

Measuring Monopsony Power

- Another measure that is common is the Herfindahl-Hirschmann Index

$$HHI = \sum_j s_j^2$$

- Where s_j is the share of employment in firm j .
- For example, if there are three firms in the market with share (0.1, 0.1 and 0.8)

$$HHI = \left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)^2 + \left(\frac{8}{10}\right)^2 = 0.66$$

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Herfindahl-Hirschmann Index (HHI)

- HHIs are common in antitrust (i.e. when two firms want to merge, they notify the Department of Justice/Federal Trade Commission)
- One way to evaluate potential anticompetitive effects of proposed merger is to compute the predicted change in concentration
- For example, if there are three firms in the market with shares (0.1, 0.1 and 0.8), and the two firms with 10% market share merge, the HHI goes from 0.66 to

$$HHI = \left(\frac{2}{10}\right)^2 + \left(\frac{8}{10}\right)^2 = 0.68$$

Sources of Monopsony Power

- We have assumed the labor supply curve to the firm is upward sloping, but why?
- Intuitively, you should think about what limits workers from moving across jobs
- If workers can seamlessly move across jobs, then that will limit the scope of monopsony power

Three Sources of Monopsony Power



Search frictions – It takes time and effort to find a job. It's not realistic that you can immediately find an equally-attractive job immediately if your firm doesn't match wage growth of the economy



Job Differentiation – This is similar to monopolistic competition. Workers like different firms for different reasons (close to home, work culture, etc.) This variation will give firms some leverage to lower wages beneath marginal revenue product of labor



Concentration/Collusion – As Adam Smith discussed in *Wealth of Nations*. Firms have an incentive not raise the wage of labor. There have been a few collusion cases we will discuss in the next section, but even if there is not outright collusion, a small number of firms facilitates market power

Why Does This Matter?

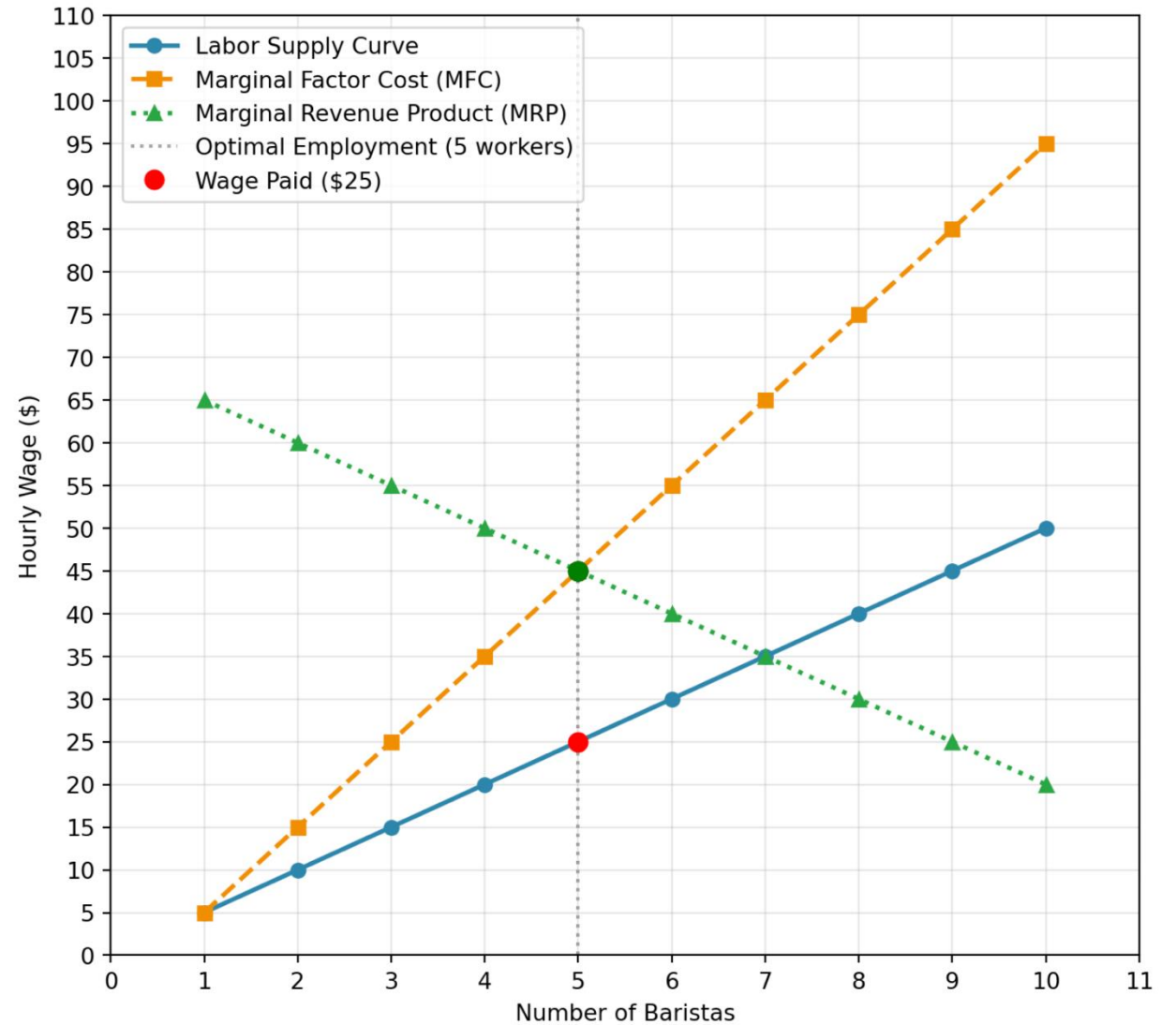
- Whether monopsony power is a more accurate model of the labor market has important implications
 - Employment is lower than optimal
 - Wages are lower than they would be in a perfectly competitive market
- Also dramatically changes implications of some policies, such as the minimum wage

Minimum Wage

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 - Employment is lower than optimal
 - Wages are lower than they would be in a perfectly competitive market
- Also dramatically changes implications of some policies, such as the minimum wage

Minimum Wage

- Let's revisit our equilibrium
- Imagine we institute a minimum wage of \$35
- How will this impact the marginal factor cost (MFC)?





Minimum Wage

- What is the marginal factor cost of the 1st worker?
- Need to pay minimum wage, so it is \$35 instead of \$5
- Let's add this point to the graph

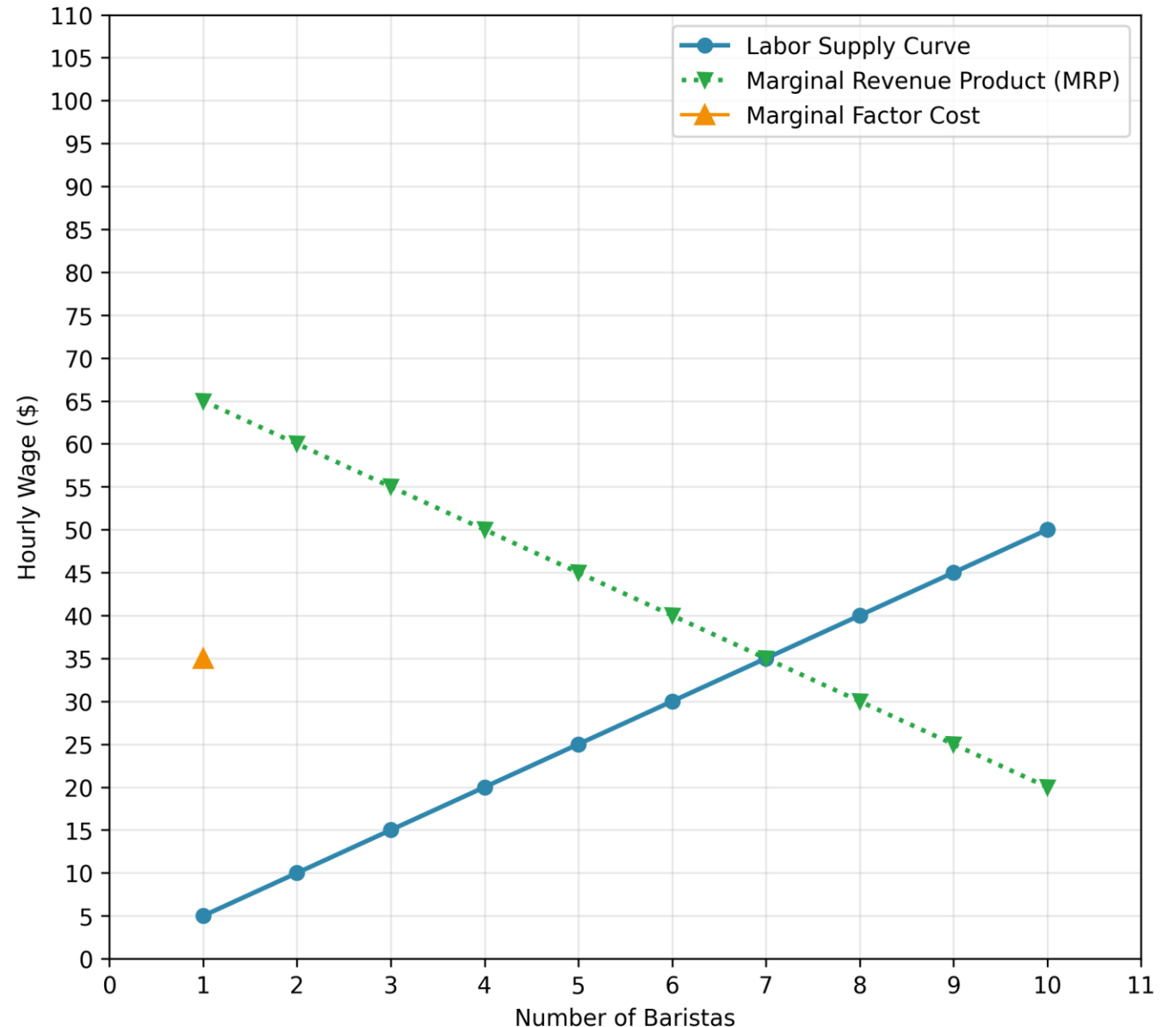


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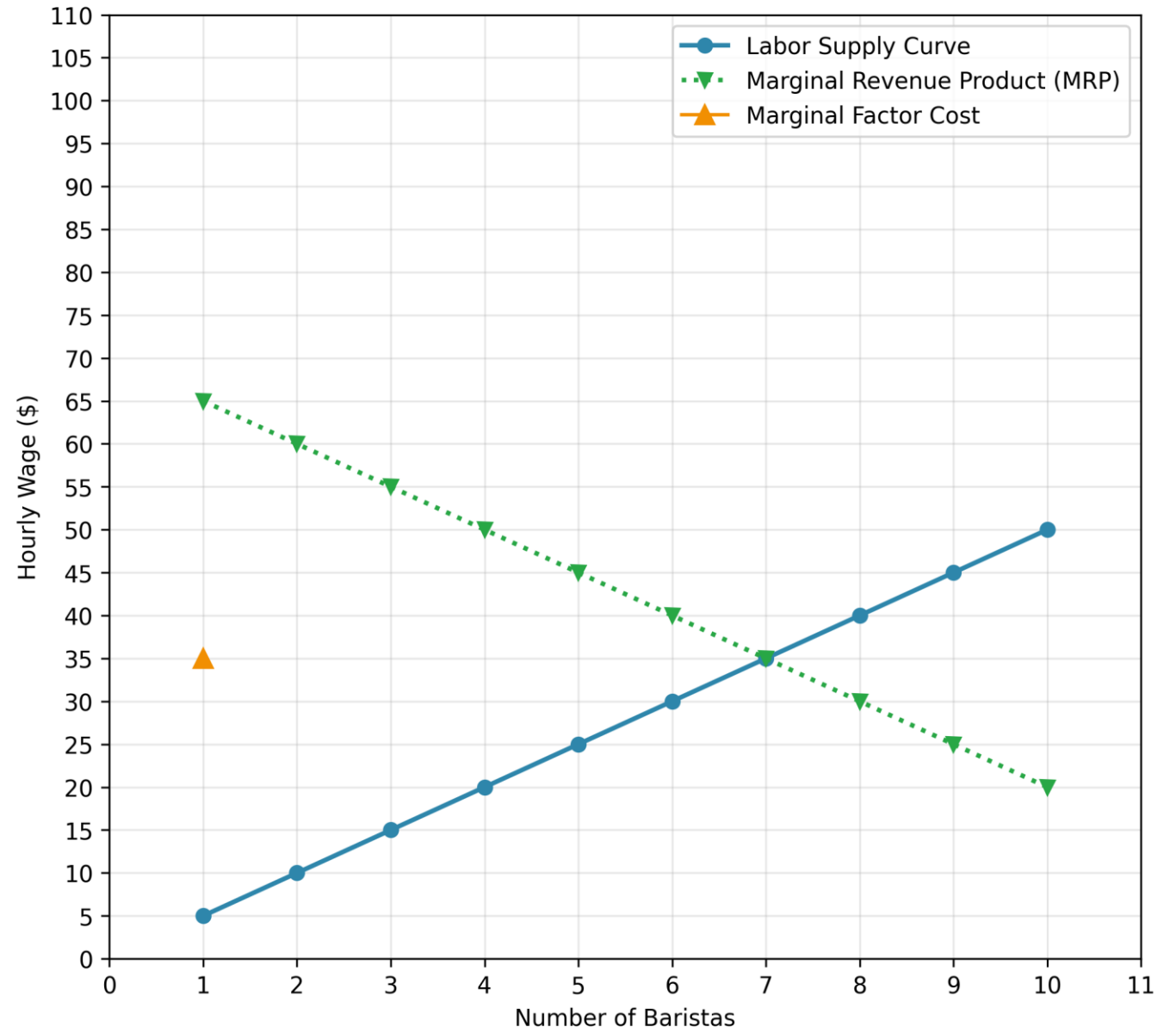
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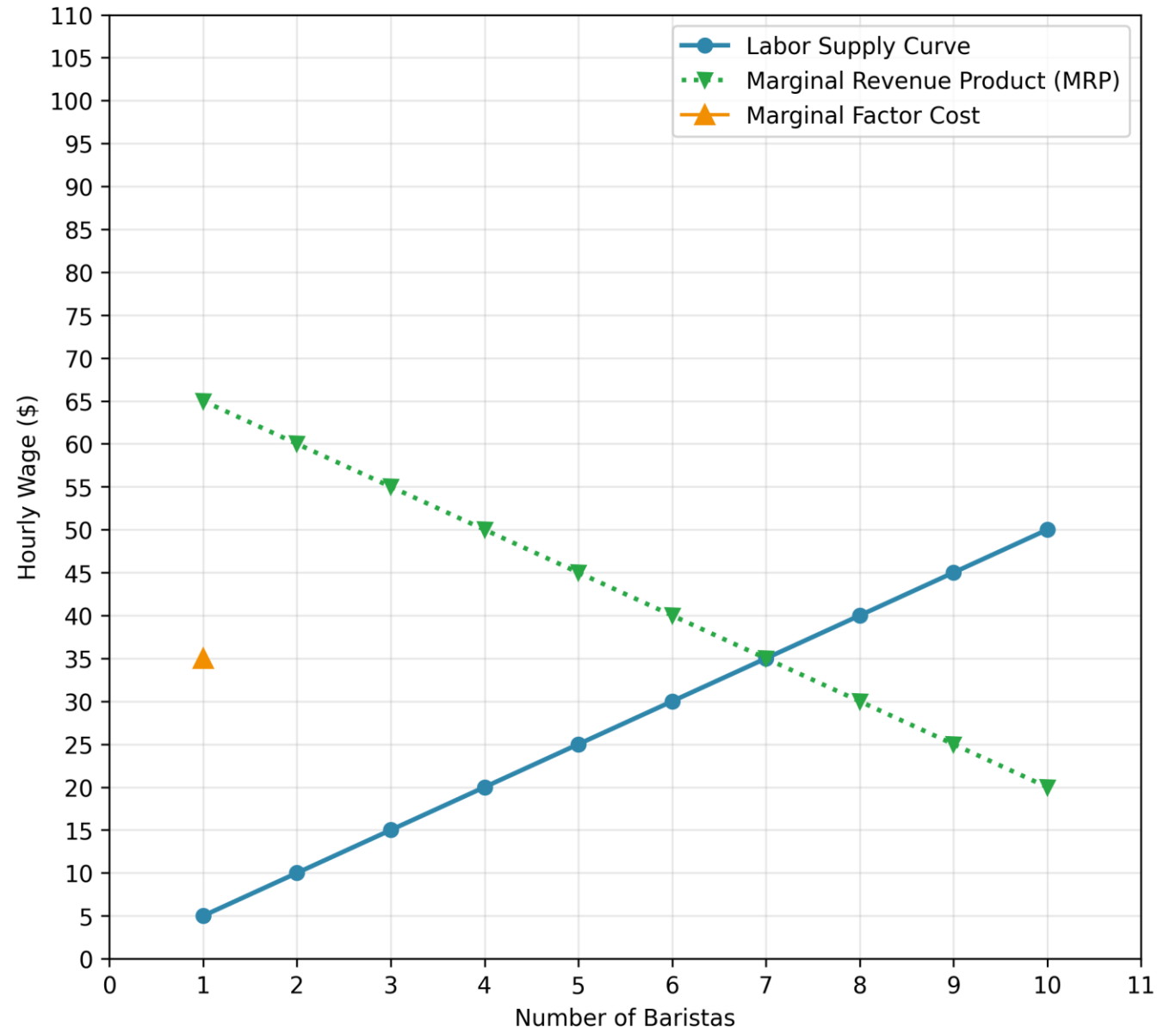
Minimum Wage

- What is the marginal factor cost of the 2nd worker?
- Need to pay minimum wage, so it is \$35 again
- Don't need to increase wage of 1st worker, you were already paying \$35



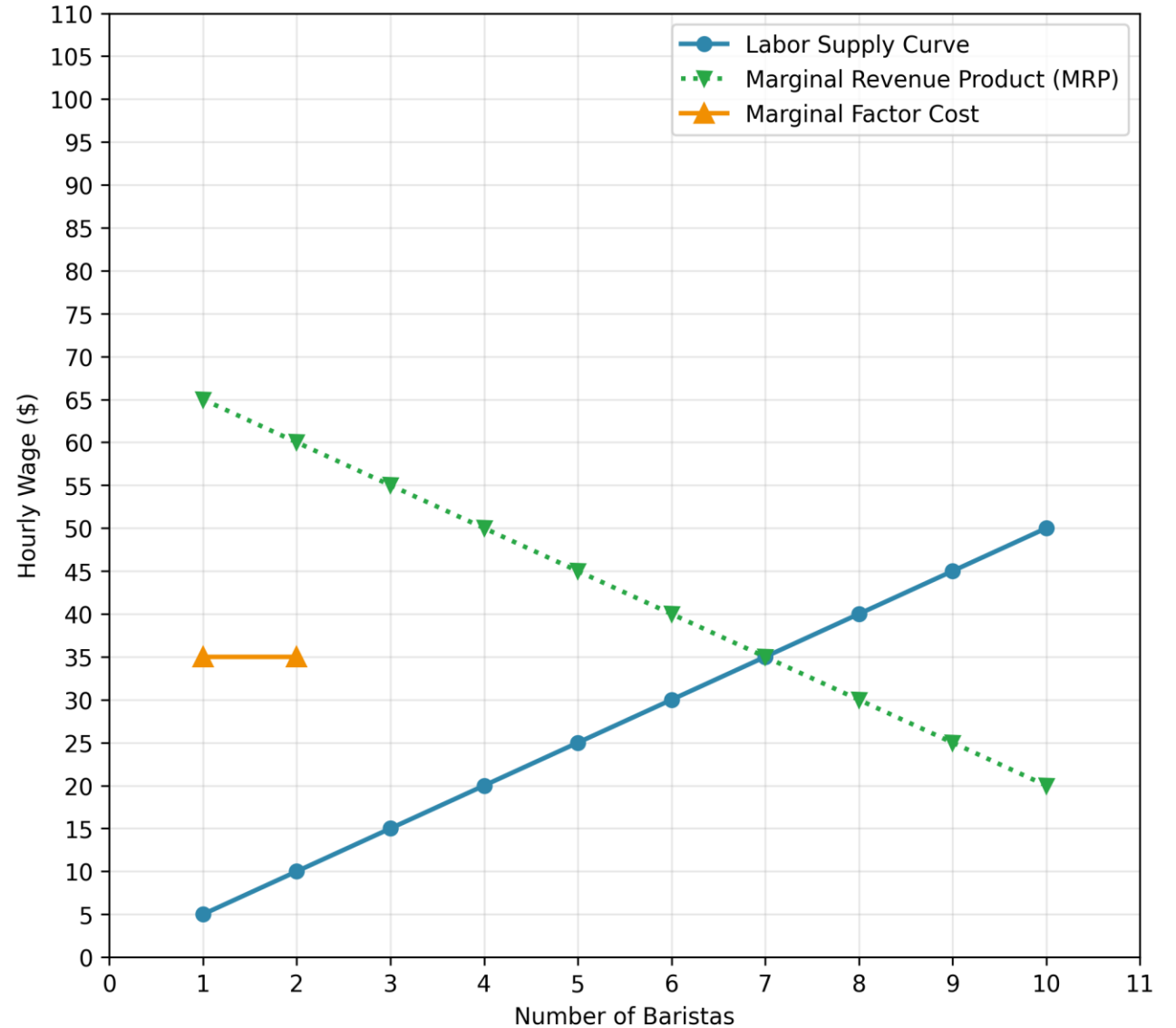
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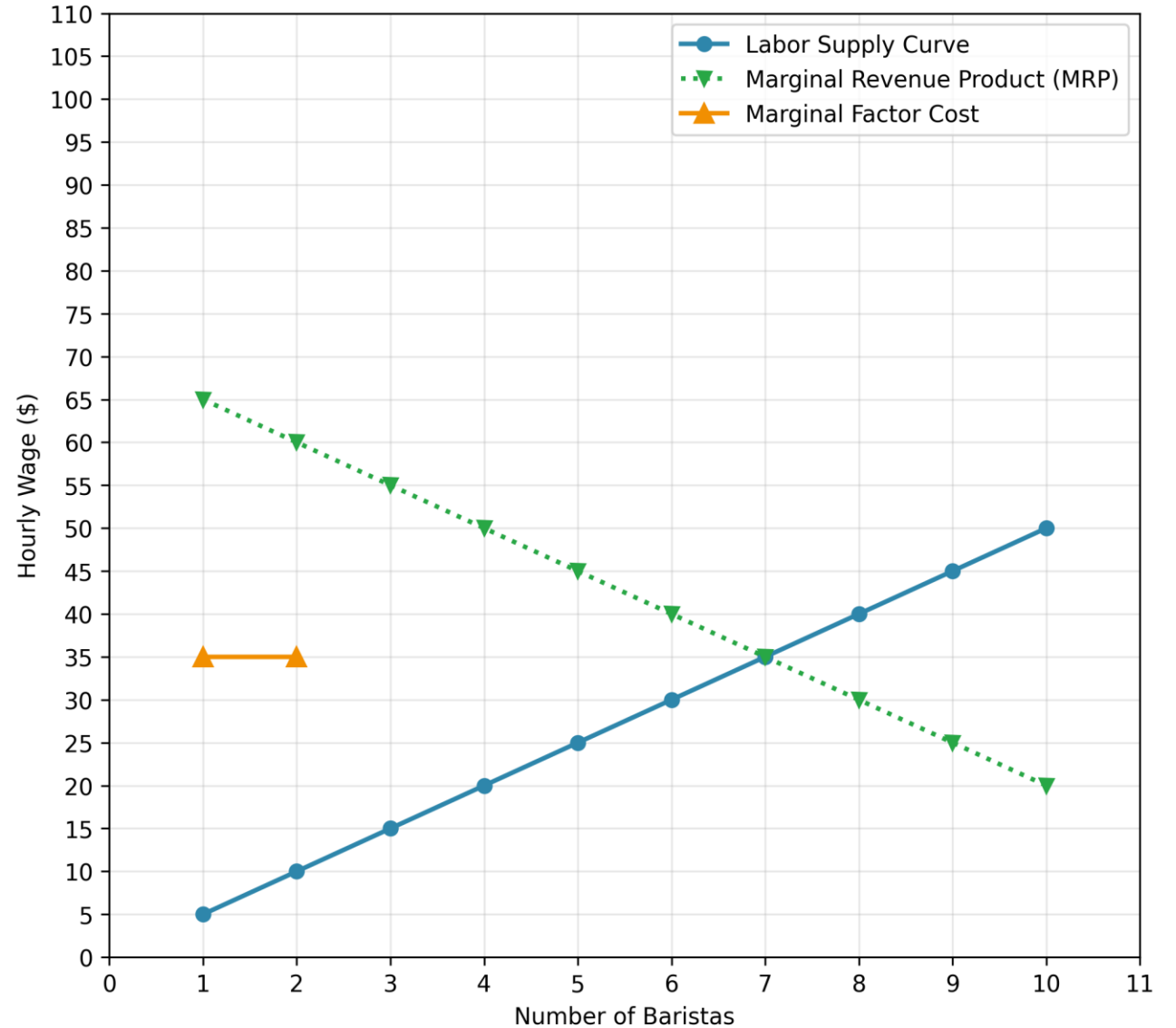
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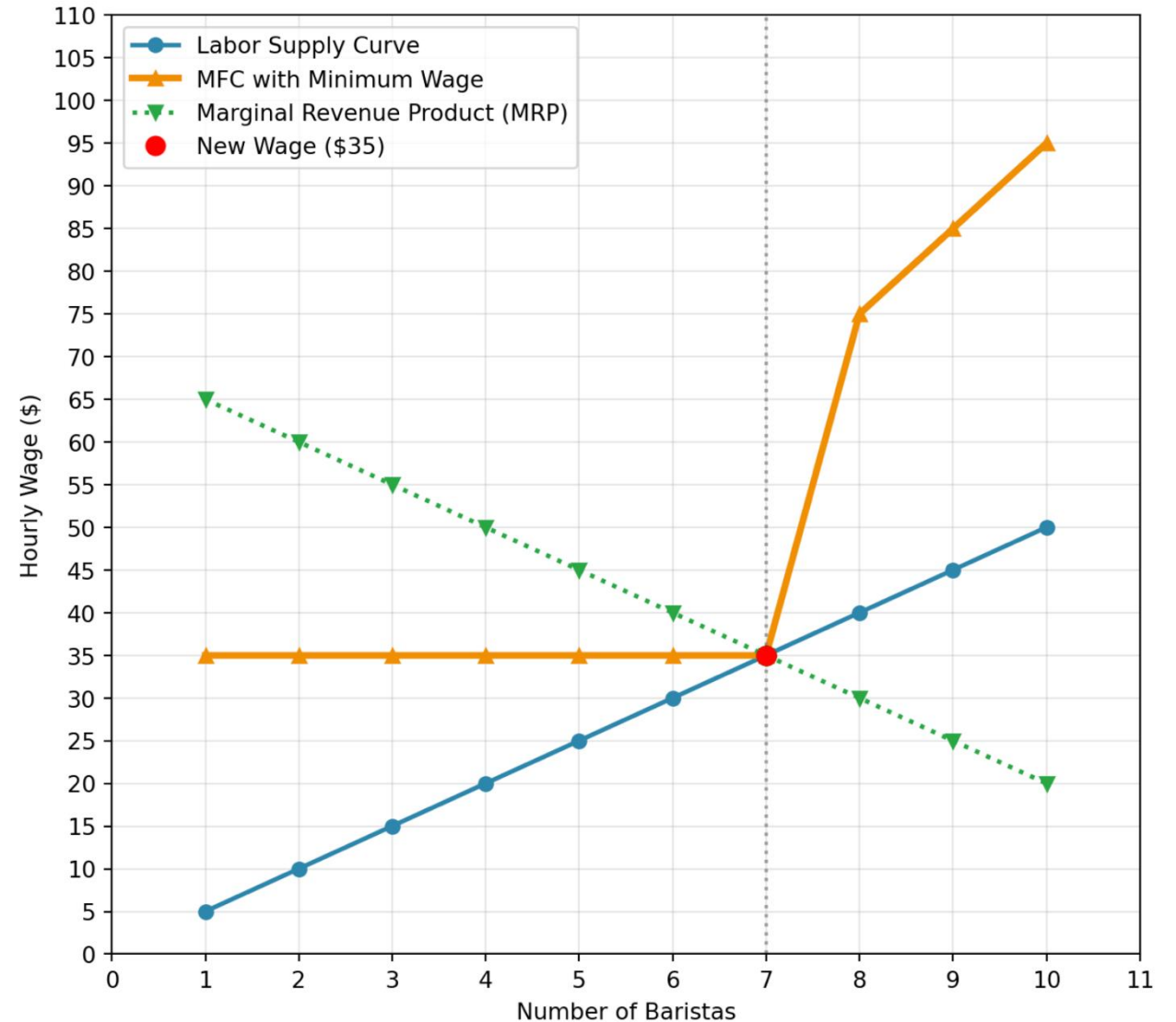
Minimum Wage

- Marginal factor cost is flat for first 7 workers (because you can attract them with a wage of \$35 or less)
- Then, to attract 8th worker need to pay \$40 to that worker, and an additional 5 to first 7 workers



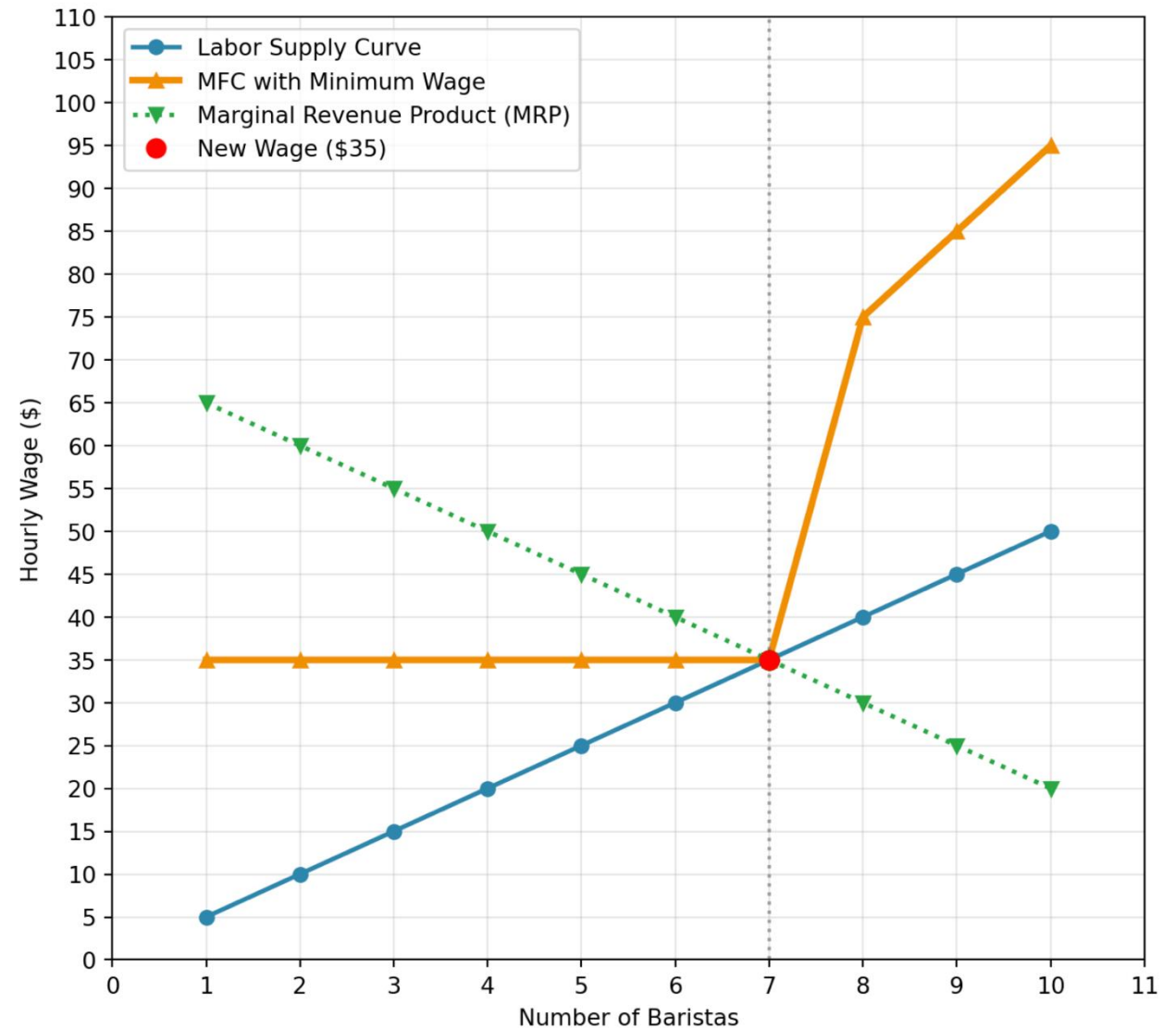
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Minimum Wage

- Counterintuitive result: a minimum wage actually *increases* employment
- Why?
- Firms take into account inframarginal costs in monopsonistic labor markets. The minimum wage can eliminate these by forcing firms to pay a higher wage to the initial workers



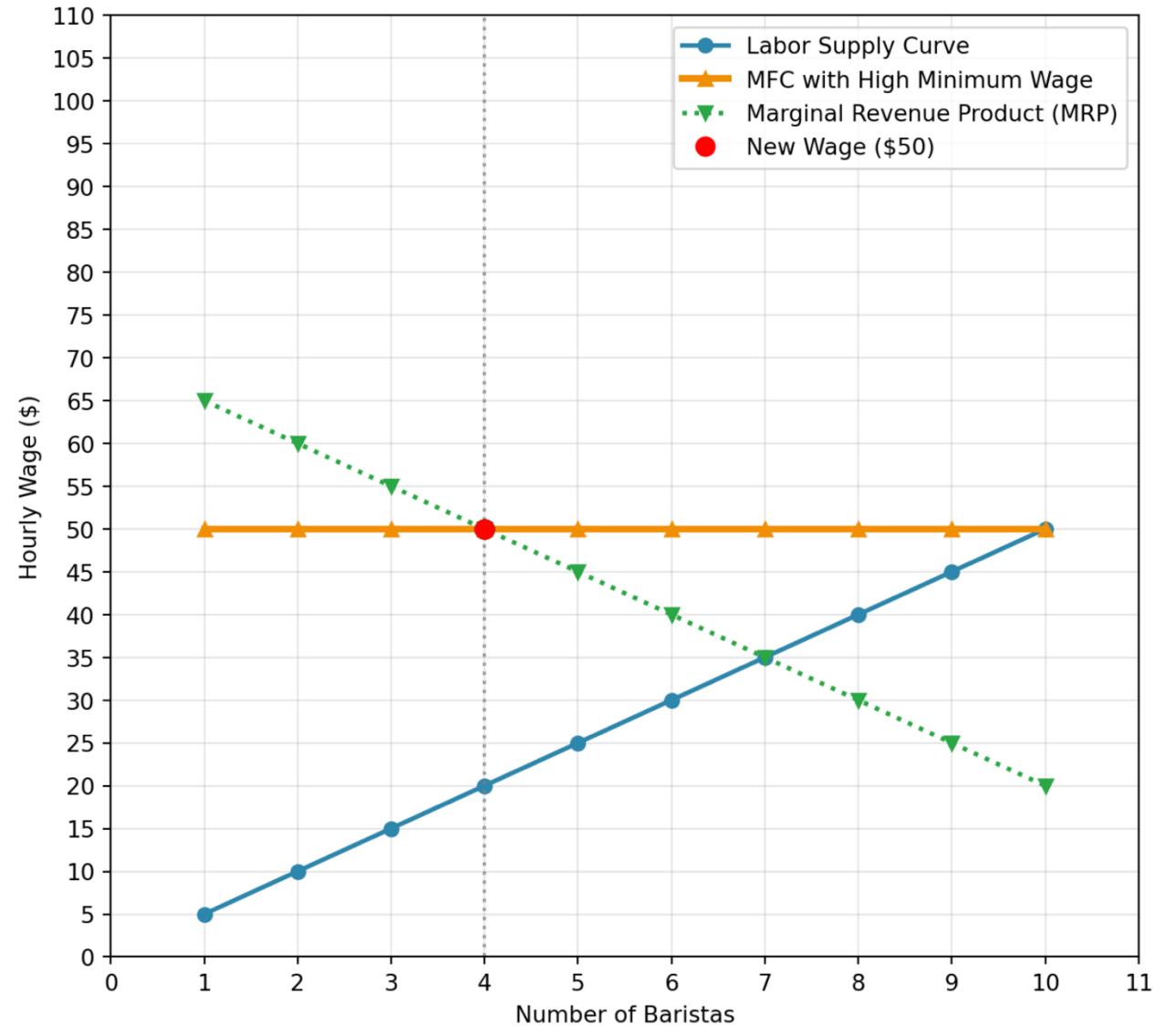


Minimum Wage

- Of course there is a point in which the wage becomes “too high”
- Imagine we set the minimum wage to \$50

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Summary

- Monopsony power is a different model of the labor market
- Many realistic features of the labor market, search frictions, job differentiation, and concentration may cause monopsony power
- Important to consider different models of the labor. They can have very different predictions with respect to policy